Data-606-Lab 6 - Inference for categorical data

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Load packages

In this lab, we will explore and visualize the data using the tidyverse suite of packages, and perform statistical inference using infer. The data can be found in the companion package for OpenIntro resources, openintro.

Let's load the packages.

The data

You will be analyzing the same dataset as in the previous lab, where you delved into a sample from the Youth Risk Behavior Surveillance System (YRBSS) survey, which uses data from high schoolers to help discover health patterns. The dataset is called yrbss.

Exercise 1

What are the counts within each category for the amount of days these students have texted while driving within the past 30 days?

Answer:

```
yrbss |>
 count(text_while_driving_30d, sort=TRUE)
```

```
## # A tibble: 9 x 2
##   text_while_driving_30d     n
##   <chr>                  <int>
## 1 0                       4792
## 2 did not drive           4646
## 3 1-2                      925
## 4 <NA>                     918
## 5 30                       827
## 6 3-5                      493
## 7 10-19                    373
## 8 6-9                      311
## 9 20-29                    298
```

Exercise 2

What is the proportion of people who have texted while driving every day in the past 30 days and never wear helmets?

Answer:

```
helms <- yrbss |>
  select(text_while_driving_30d, helmet_12m) |>
  filter(
    !is.na(text_while_driving_30d),
    helmet_12m == "never"
  ) |>
  mutate(
    text = ifelse(text_while_driving_30d == "30",
                  "everyday", "other")
  ) |>
  count(text)

helms
```

```
## # A tibble: 2 x 2
##   text         n
##   <chr>    <int>
## 1 everyday   463
## 2 other     6040
```

```
# Calculate the proportion who texted every day
proportion <- helms$n[helms$text == "everyday"] / sum(helms$n)

proportion
```

```
## [1] 0.07119791
```

The output is approximately 0.0712 , or 7.12% .

Among students who never wear helmets, 463 reported texting while driving every day during the past 30 days, while 6,040 reported other responses. Therefore, the proportion who texted while driving every day was 463 / 6,503 = 0.0712, or approximately 7.12%.

Remember that you can use filter to limit the dataset to just non-helmet wearers. Here, we will name the dataset no_helmet.

```
data('yrbss', package='openintro')
no_helmet <- yrbss %>%
  filter(helmet_12m == "never")
```

Also, it may be easier to calculate the proportion if you create a new variable that specifies whether the individual has texted every day while driving over the past 30 days or not. We will call this variable text_ind.

```
no_helmet <- no_helmet %>%
  mutate(text_ind = ifelse(text_while_driving_30d == "30", "yes", "no"))
```

Inference on proportions

When summarizing the YRBSS, the Centers for Disease Control and Prevention seeks insight into the population parameters. To do this, you can answer the question, "What proportion of people in your sample

reported that they have texted while driving each day for the past 30 days?" with a statistic; while the question "What proportion of people on earth have texted while driving each day for the past 30 days?" is answered with an estimate of the parameter.

The inferential tools for estimating population proportion are analogous to those used for means in the last chapter: the confidence interval and the hypothesis test.

```
set.seed(1234)

no_helmet %>%
  filter(!is.na(text_ind)) %>% # filter out na's
  specify(response = text_ind, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1   0.0652   0.0777
```

Note that since the goal is to construct an interval estimate for a proportion, it's necessary to both include the success argument within specify, which accounts for the proportion of non-helmet wearers than have consistently texted while driving the past 30 days, in this example, and that stat within calculate is here "prop", signaling that you are trying to do some sort of inference on a proportion.

Exercise 3

What is the margin of error for the estimate of the proportion of non-helmet wearers that have texted while driving each day for the past 30 days based on this survey?

Answer:

The margin of error is:

$$ME = \frac{Upper\ CI - Lower\ CI}{2}$$

$$ME = \frac{0.0777 - 0.0652}{2} = 0.00625$$

```
lower_ci <- 0.0652
upper_ci <- 0.0777

ME <- (upper_ci - lower_ci) / 2
ME
```

```
## [1] 0.00625
```

The margin of error is approximately 0.625 percentage points.

Exercise 4

Using the infer package, calculate confidence intervals for two other categorical variables (you'll need to decide which level to call "success", and report the associated margins of error. Interpet the interval in context of the data. It may be helpful to create new data sets for each of the two countries first, and then use these data sets to construct the confidence intervals.

Answer:

```
glimpse(yrbss)
```

```
## Rows: 13,583
## Columns: 13
## $ age                     <int> 14, 14, 15, 15, 15, 15, 15, 14, 15, 15, 15, 1~
## $ gender                  <chr> "female", "female", "female", "female", "fema~
## $ grade                   <chr> "9", "9", "9", "9", "9", "9", "9", "9", "9", ~
## $ hispanic                <chr> "not", "not", "hispanic", "not", "not", "not"~
## $ race                    <chr> "Black or African American", "Black or Africa~
## $ height                  <dbl> NA, NA, 1.73, 1.60, 1.50, 1.57, 1.65, 1.88, 1~
## $ weight                  <dbl> NA, NA, 84.37, 55.79, 46.72, 67.13, 131.54, 7~
## $ helmet_12m              <chr> "never", "never", "never", "never", "did not ~
## $ text_while_driving_30d  <chr> "0", NA, "30", "0", "did not drive", "did not~
## $ physically_active_7d    <int> 4, 2, 7, 0, 2, 1, 4, 4, 5, 0, 0, 0, 4, 7, 7, ~
## $ hours_tv_per_school_day <chr> "5+", "5+", "5+", "2", "3", "5+", "5+", "5+",~
## $ strength_training_7d    <int> 0, 0, 0, 0, 1, 0, 2, 0, 3, 0, 3, 0, 0, 7, 7, ~
## $ school_night_hours_sleep <chr> "8", "6", "<5", "6", "9", "8", "9", "6", "<5"~
```

Let's calculate confidence intervals for sleep and hours of tv.

```
#students who sleep 8 hours or more
unique(yrbss$school_night_hours_sleep)
```

```
## [1] "8"   "6"   "<5"  "9"   "10+" "7"   "5"   NA
```

```
set.seed(1234)

sleep <- yrbss |>
  filter(!is.na(school_night_hours_sleep)) |>
  mutate(eight_hours = ifelse(school_night_hours_sleep %in% c("8","9","10+"), "yes", "no"))

sleep |>
  count(eight_hours) |>
  mutate(p = n / sum(n))
```

```
## # A tibble: 2 x 3
##   eight_hours     n     p
##   <chr>       <int> <dbl>
## 1 no           8564 0.694
## 2 yes          3771 0.306
```

```
sleep |>
  specify(response = eight_hours, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1    0.298    0.314
```

The 95% bootstrap confidence interval for the proportion of students who get at least 8 hours of sleep on a school night is approximately 0.298 to 0.314. This means we are 95% confident that the population proportion is between 29.8% and 31.4%.

```
# Calculate sample proportion
p <- mean(sleep$eight_hours == "yes")

# Sample size
n <- nrow(sleep)

# Z-score for 95% confidence
z <- 1.96

# Standard error
se <- sqrt(p * (1 - p) / n)

# Margin of error
me <- z * se

# Display results
p
```

```
## [1] 0.3057154
```

```
n
```

```
## [1] 12335
```

```
me
```

```
## [1] 0.008130438
```

The margin of error is 0.0081

Let's look at hours of tv:

```
tv <- yrbss |>
  filter(!is.na(hours_tv_per_school_day)) |>
  mutate(five_hours_or_more = ifelse(hours_tv_per_school_day == "5+", "yes","no"))

tv |>
  count(five_hours_or_more) |>
  mutate(p = n / sum(n))
```

```
## # A tibble: 2 x 3
##   five_hours_or_more     n     p
##   <chr>              <int> <dbl>
## 1 no                 11650 0.880
## 2 yes                 1595 0.120
```

Calculating the 95% confidence interval

```
tv |>
  specify(response = five_hours_or_more, success = "yes") %>%
  generate(reps = 1000, type = "bootstrap") %>%
  calculate(stat = "prop") %>%
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1    0.115    0.126
```

The 95% confidence interval for the probability of a student watching five or more hours of tv is 0.115-0.126.

Calculating margin of error:

```
n <- nrow(tv)
p <- sum(tv$five_hours_or_more == "yes")/n
z <- 1.96 # z-score 95%
se <- sqrt(p * (1-p) / n)
me <- z * se
me
```

```
## [1] 0.005542701
```

The corrected TV margin of error is approximately 0.00554, or 0.554 percentage points.

How does the proportion affect the margin of error?

The difference between the upper CI and lower CI is 0.011, so the margin of error is nearly equal to the distance between the upper and lower CIs.

Imagine you've set out to survey 1000 people on two questions: are you at least 6-feet tall? and are you left-handed? Since both of these sample proportions were calculated from the same sample size, they should have the same margin of error, right? Wrong! While the margin of error does change with sample size, it is also affected by the proportion.

Think back to the formula for the standard error: $SE=\sqrt{p(1-p)/n}$. This is then used in the formula for the margin of error for a 95% confidence interval:

$ME=1.96\times SE=1.96\times\sqrt{p(1-p)/n}$. Since the population proportion p is in this ME formula, it should make sense that the margin of error is in some way dependent on the population proportion. We can visualize this relationship by creating a plot of ME vs. p .

Since sample size is irrelevant to this discussion, let's just set it to some value (n=1000) and use this value in the following calculations:

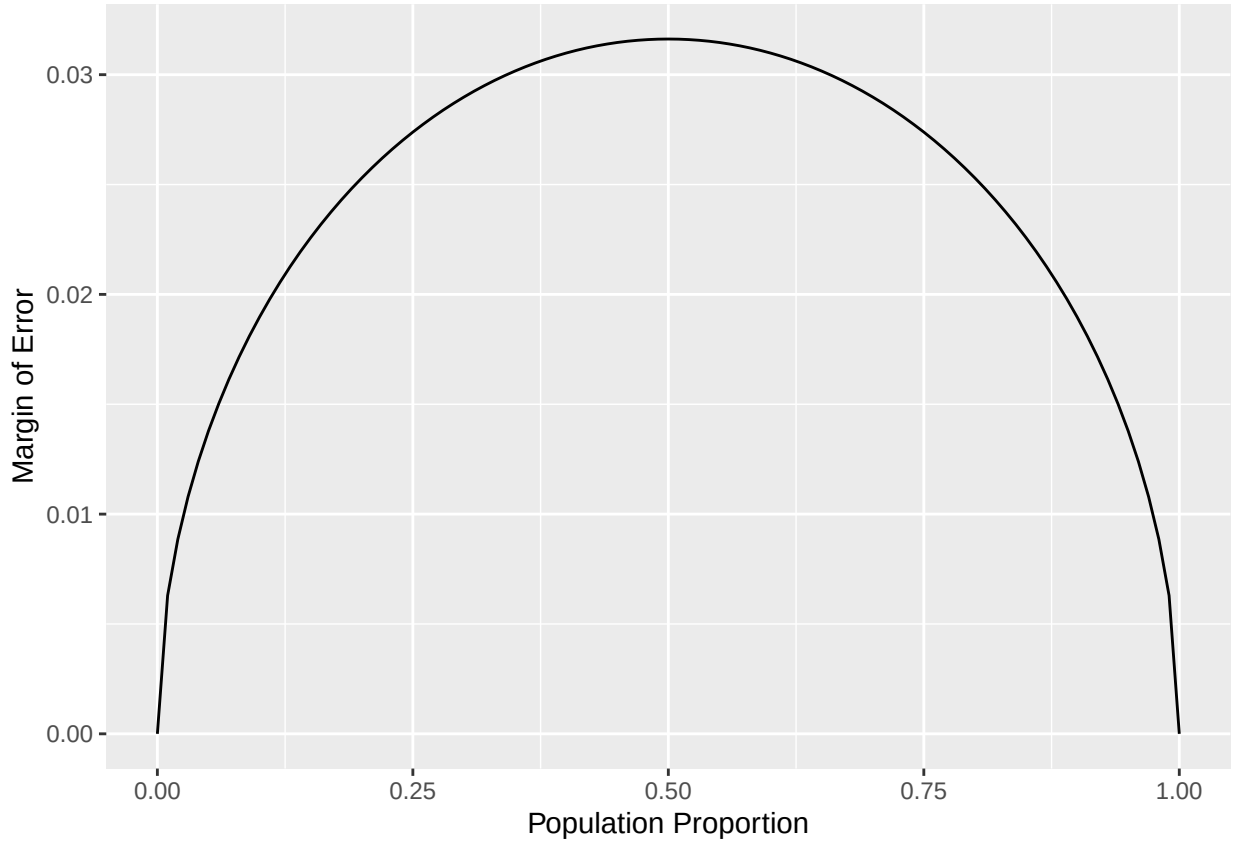
```r
n <- 1000
```

The first step is to make a variable p that is a sequence from 0 to 1 with each number incremented by 0.01. You can then create a variable of the margin of error (me) associated with each of these values of p using the familiar approximate formula (ME=2×SE).

```r
p <- seq(from = 0, to = 1, by = 0.01)
me <- 2 * sqrt(p * (1 - p)/n)
```

Lastly, you can plot the two variables against each other to reveal their relationship. To do so, we need to first put these variables in a data frame that you can call in the ggplot function.

```r
dd <- data.frame(p = p, me = me)
ggplot(data = dd, aes(x = p, y = me)) +
  geom_line() +
  labs(x = "Population Proportion", y = "Margin of Error")
```



Exercise 5

Describe the relationship between p and me. Include the margin of error vs. population proportion plot you constructed in your answer. For a given sample size, for which value of p is margin of error maximized?

Answer:

The margin of error increases as the population proportion approaches 0.5 and decreases as the proportion approaches either 0 or 1. For a fixed sample size, the margin of error is maximized at p = 0.5 because the variability p(1 − p) is greatest at this value.

Success-failure condition

We have emphasized that you must always check conditions before making inference. For inference on proportions, the sample proportion can be assumed to be nearly normal if it is based upon a random sample of independent observations and if both np 10 and n(1−p) 10 . This rule of thumb is easy enough to follow, but it makes you wonder: what's so special about the number 10?

The short answer is: nothing. You could argue that you would be fine with 9 or that you really should be using 11. What is the "best" value for such a rule of thumb is, at least to some degree, arbitrary. However, when np and n(1−p) reaches 10 the sampling distribution is sufficiently normal to use confidence intervals and hypothesis tests that are based on that approximation.

You can investigate the interplay between n and p and the shape of the sampling distribution by using simulations. Play around with the following app to investigate how the shape, center, and spread of the distribution of p^ changes as n and p changes.

Exercise 6

Describe the sampling distribution of sample proportions at n=300 and p=0.1. Be sure to note the center, spread, and shape.

Answer:

The sampling distribution is approximately normal and centered at p = 0.1. The standard error is approximately 0.0173, indicating a relatively small spread. Most sample proportions fall approximately between 0.066 and 0.134.

Exercise 7

Keep n constant and change p. How does the shape, center, and spread of the sampling distribution vary as p changes. You might want to adjust min and max for the x-axis for a better view of the distribution.

Answer:

As the population proportion p increases, the center of the sampling distribution also increases because the mean of the sample proportions is p. The spread increases as p approaches 0.5 and decreases as p approaches 0 or 1. The shape of the sampling distribution is approximately symmetric and bell-shaped when the sample size is sufficiently large and p is not too close to 0 or 1. When p is close to 0 or 1, the distribution may be skewed, especially for smaller sample sizes.

Exercise 8

Now also change n. How does n appear to affect the distribution of p^?

Answer:

As the sample size n increases, the spread of the sampling distribution of $\hat{p}$ decreases because the standard error becomes smaller. The center remains the same at the population proportion p. The distribution also becomes more approximately normal as the sample size increases, provided the population proportion is not too close to 0 or 1.

More Practice

For some of the exercises below, you will conduct inference comparing two proportions. In such cases, you have a response variable that is categorical, and an explanatory variable that is also categorical, and you are comparing the proportions of success of the response variable across the levels of the explanatory variable. This means that when using infer, you need to include both variables within specify.

Exercise 9

Is there convincing evidence that those who sleep 10+ hours per day are more likely to strength train every day of the week? As always, write out the hypotheses for any tests you conduct and outline the status of the conditions for inference. If you find a significant difference, also quantify this difference with a confidence interval.

Answer:

Hypotheses

Null hypothesis (H_0): The proportion of students who strength train every day is the same for students who sleep 10+ hours and students who sleep less than 10 hours.

Alternative hypothesis (H_A): The proportion of students who strength train every day is higher among students who sleep 10+ hours than among students who sleep less than 10 hours.

```
library(tidyverse)
library(infer)

# Prepare the data
exercise9 <- yrbss |>
  filter(
    !is.na(school_night_hours_sleep),
    !is.na(strength_training_7d)
  ) |>
  mutate(
    sleep_group = ifelse(
      school_night_hours_sleep == "10+",
      "10+ hours",
      "Less than 10 hours"
    ),
    strength_daily = ifelse(
      strength_training_7d == 7,
      "yes",
      "no"
    )
  )

# Count students in each group
exercise9 |>
  count(sleep_group, strength_daily)
```

```
## # A tibble: 4 x 3
##   sleep_group        strength_daily     n
##   <chr>              <chr>          <int>
## 1 10+ hours          no               228
## 2 10+ hours          yes               84
```

```
## 3 Less than 10 hours no              9949
## 4 Less than 10 hours yes             1958
```

```r
# Calculate the observed difference in proportions
obs_diff <- exercise9 |>
  specify(
    response = strength_daily,
    explanatory = sleep_group,
    success = "yes"
  ) |>
  calculate(
    stat = "diff in props",
    order = c("10+ hours", "Less than 10 hours")
  )

obs_diff
```

```
## Response: strength_daily (factor)
## Explanatory: sleep_group (factor)
## # A tibble: 1 x 1
##    stat
##   <dbl>
## 1 0.105
```

```r
# Generate the null distribution
set.seed(1234)

null_dist <- exercise9 |>
  specify(
    response = strength_daily,
    explanatory = sleep_group,
    success = "yes"
  ) |>
  hypothesize(null = "independence") |>
  generate(reps = 1000, type = "permute") |>
  calculate(
    stat = "diff in props",
    order = c("10+ hours", "Less than 10 hours")
  )

# Calculate the p-value
p_value <- null_dist |>
  get_p_value(
    obs_stat = obs_diff$stat,
    direction = "greater"
  )
```

```
## Warning: Please be cautious in reporting a p-value of 0. This result is an approximation
## based on the number of `reps` chosen in the `generate()` step.
## i See `get_p_value()` (`?infer::get_p_value()`) for more information.
```

```
p_value
```

```
## # A tibble: 1 x 1
##   p_value
##     <dbl>
## 1       0
```

```
set.seed(1234)

boot_dist <- exercise9 |>
  specify(
    response = strength_daily,
    explanatory = sleep_group,
    success = "yes"
  ) |>
  generate(reps = 1000, type = "bootstrap") |>
  calculate(
    stat = "diff in props",
    order = c("10+ hours", "Less than 10 hours")
  )

boot_dist |>
  get_ci(level = 0.95)
```

```
## # A tibble: 1 x 2
##   lower_ci upper_ci
##      <dbl>    <dbl>
## 1   0.0579    0.154
```

Statistical Results

Among students who reported sleeping 10 or more hours per school night, 84 out of 312 (26.92%) reported strength training every day. Among students who reported sleeping less than 10 hours, 1,958 out of 11,907 (16.44%) reported strength training every day.

The observed difference in proportions was approximately 0.105, or 10.5 percentage points.

A permutation test with 1,000 repetitions produced a p-value below 0.001. Since the p-value is less than the significance level of 0.05, we reject the null hypothesis.

The 95% bootstrap confidence interval for the difference in proportions was 0.0579 to 0.154, or approximately 5.79 to 15.4 percentage points.

Conditions for Inference

The success-failure condition is satisfied because both groups contain at least 10 successes and 10 failures. The independence condition is assumed for this introductory analysis, although the survey's sampling design may affect this assumption.

Conclusion

There is statistically significant evidence that students who sleep 10 or more hours per school night are more likely to report strength training every day than students who sleep less than 10 hours. The difference is approximately 10.5 percentage points, and the 95% confidence interval supports a positive association between longer sleep and daily strength training. These findings indicate an association, not a causal relationship.

Exercise 10

Let's say there has been no difference in likeliness to strength train every day of the week for those who sleep 10+ hours. What is the probablity that you could detect a change (at a significance level of 0.05) simply by chance? Hint: Review the definition of the Type 1 error.

Answer:

If there is truly no difference between the two groups, the probability of making a Type I error is 0.05, or 5%, when the significance level is 0.05. A Type I error occurs when we reject the null hypothesis even though it is true.

Exercise 11

Suppose you're hired by the local government to estimate the proportion of residents that attend a religious service on a weekly basis. According to the guidelines, the estimate must have a margin of error no greater than 1% with 95% confidence. You have no idea what to expect for p. How many people would you have to sample to ensure that you are within the guidelines? Hint: Refer to your plot of the relationship between p and margin of error. This question does not require using a dataset.

Answer:

```
ME <- 0.01 #margin of error
z <- 1.96 #z-score for 95% confidence

p <- 0.5 #margin for error is highest at .5 of the population proportion

n <- z**2 * p*(1-p)/ME**2
n
```

```
## [1] 9604
```

At least 9604 people